

Consumer Shopping Behavior Analysis

Data Analytics Portfolio Project | Power BI

Dataset	Consumer Shopping Behavior Survey
Records	500 respondents
Tools	Power BI Power Query DAX
Analysis	Descriptive & Exploratory Analysis

Report Contents

- 1. Project Overview
- 2. Business Problem
- 3. Dataset Description
- 4. Data Cleaning & Transformation
- 5. Exploratory Data Analysis
- 6. Power BI Dashboard
- 7. Key Insights
- 8. Business Recommendations
- 9. Tools & Technologies
- 10. Limitations
- 11. Conclusion

1. Project Overview

This project analyzes consumer shopping behavior using survey data from 500 respondents. The analysis focuses on shopping preferences, online shopping habits, purchase decision factors, spending patterns, payment preferences, and customer experiences with online shopping.

The dataset was cleaned and transformed using Power Query, explored through multiple analytical views, and visualized using an interactive Power BI dashboard. The dashboard supports filtering by gender, age group, income range, and country so that consumer behavior can be explored across demographic segments.

The objective is to convert raw survey data into clear, decision-oriented insights that may support pricing, promotion, customer experience, payment, and omnichannel retail decisions.

Item	Details
Project	Consumer Shopping Behavior Analysis
Dataset	Consumer Shopping Behavior Survey
Records	500 respondents
Tools	Power BI, Power Query, DAX
Analysis Type	Descriptive & Exploratory Analysis
Final Output	Interactive Power BI Dashboard

2. Business Problem

Retail and e-commerce businesses need to understand not only where consumers shop, but also what influences their purchasing decisions, how they prefer to pay, how much they spend, and how they experience online shopping. This project uses survey responses to identify patterns that can help frame customer-focused business decisions.

Analysis Objectives

- Identify the shopping channels consumers prefer.
- Understand which factors are most strongly reported as purchase drivers.

- Measure online shopping frequency and preferred shopping platforms/devices.
- Examine non-essential spending bands and leading spending categories.
- Understand preferred payment methods and BNPL usage.
- Assess online shopping satisfaction, trust in reviews, return behavior, and purchase regret.
- Explore selected relationships between behavioral variables without claiming causation.

3. Dataset Description

The dataset contains 500 survey responses across 30 fields. It combines demographic attributes with shopping behavior, purchase drivers, spending and payment preferences, and online shopping experience measures.

Data Area	Example Fields
Demographics	Age, Gender, Occupation Status, Country/Region, Monthly Income Range
Shopping Behavior	Shop Most Where, Online Shopping Frequency, Preferred Platform, Primary Device
Purchase Drivers	Review Influence, Social Ads Influence, Price Comparison, Discount Importance, Likelihood After Ad, Impulse Purchase
Spending & Payment	Top Spending Category, Monthly Non-Essential Spend, Preferred Payment Method, BNPL Usage
Customer Experience	Online Satisfaction, Trust in Reviews, Return Frequency, Purchase Regret
Preference Reasons	Reason Prefer Online, Reason Prefer In-Store, Best Shopping Mode

Most purchase-driver scores use a 1–5 scale, while online-shopping satisfaction and trust in online reviews use 1–10 scales. Monthly spending and income are provided as bands rather than exact monetary values.

4. Data Cleaning & Transformation

Data preparation was completed in Power Query before analysis. A raw query was retained and a referenced cleaned query was used for transformation so that the original data remained available.

- Applied Trim and Clean to text/categorical fields to remove unnecessary spaces and non-printing characters.
- Standardized missing categorical responses to 'Unknown' where appropriate, while retaining missing numeric survey scores as null rather than replacing them with zero.
- Converted the timestamp using the appropriate day/month/year locale and retained it as Date/Time.
- Cleaned the Age field by converting valid ages to whole numbers and replacing invalid text entries (for example test strings or names) with null.
- Created an Age Group field for analysis and filtering.
- Validated numeric score columns and retained their intended 1–5 or 1–10 ranges.
- Created ordinal/helper fields for Online Shopping Frequency, Return Frequency, Shopping List behavior, and BNPL usage.
- Created Spend_Band_Sort and Income_Sort helper columns so categorical ranges display in logical order rather than alphabetically.
- Checked duplicate rows and Response_ID uniqueness; no complete duplicate rows and no duplicate Response_IDs were identified.
- Retained all useful survey fields during EDA rather than removing columns prematurely.

Examples of Data Quality Issues Addressed

Issue	Treatment
Invalid Age text	Converted valid ages to whole numbers; invalid text became null
Missing categorical responses	Used 'Unknown' where useful for transparent reporting
Missing numeric scores	Kept as null so missing responses were not treated as zero
Mixed timestamp format	Converted with locale-aware Date/Time transformation
Categorical sort order	Created numeric helper columns for income, spend, frequency, returns, and BNPL
Text inconsistencies	Applied Trim/Clean and standardized capitalization where required

5. Exploratory Data Analysis

EDA was performed across consumer profile, shopping behavior, purchase drivers, spending/payment behavior, and customer experience. Descriptive charts were complemented by selected relationship analyses.

Selected Relationship Findings

- Social-ad influence vs likelihood after an ad: purchase likelihood generally increased as reported social-ad influence increased (approximately 2.27 at influence score 1 to 3.65 at score 5).
- Discount importance vs price comparison: average price-comparison scores remained high and fairly flat across discount-importance levels, showing no clear increasing pattern.
- BNPL usage vs monthly non-essential spending: spending-band composition did not show a simple progression toward higher spending as BNPL usage increased.
- Review trust vs online satisfaction: average satisfaction remained relatively stable across review-trust scores, with no obvious strong relationship.
- Return frequency vs purchase regret: average regret scores were similar across Never, Sometimes, and Often return groups.
- Shopping preference vs online satisfaction: satisfaction was broadly similar across shopping-preference groups; in-store-focused respondents were slightly higher in this sample.

Because this is survey/observational data, the analysis describes patterns and associations. It does not establish that one behavior causes another.

6. Power BI Dashboard

The final dashboard summarizes the most decision-relevant findings and allows interactive filtering by Gender, Age Group, Income Range, and Country. The dashboard intentionally uses a limited number of visuals to keep the presentation readable and executive-friendly.

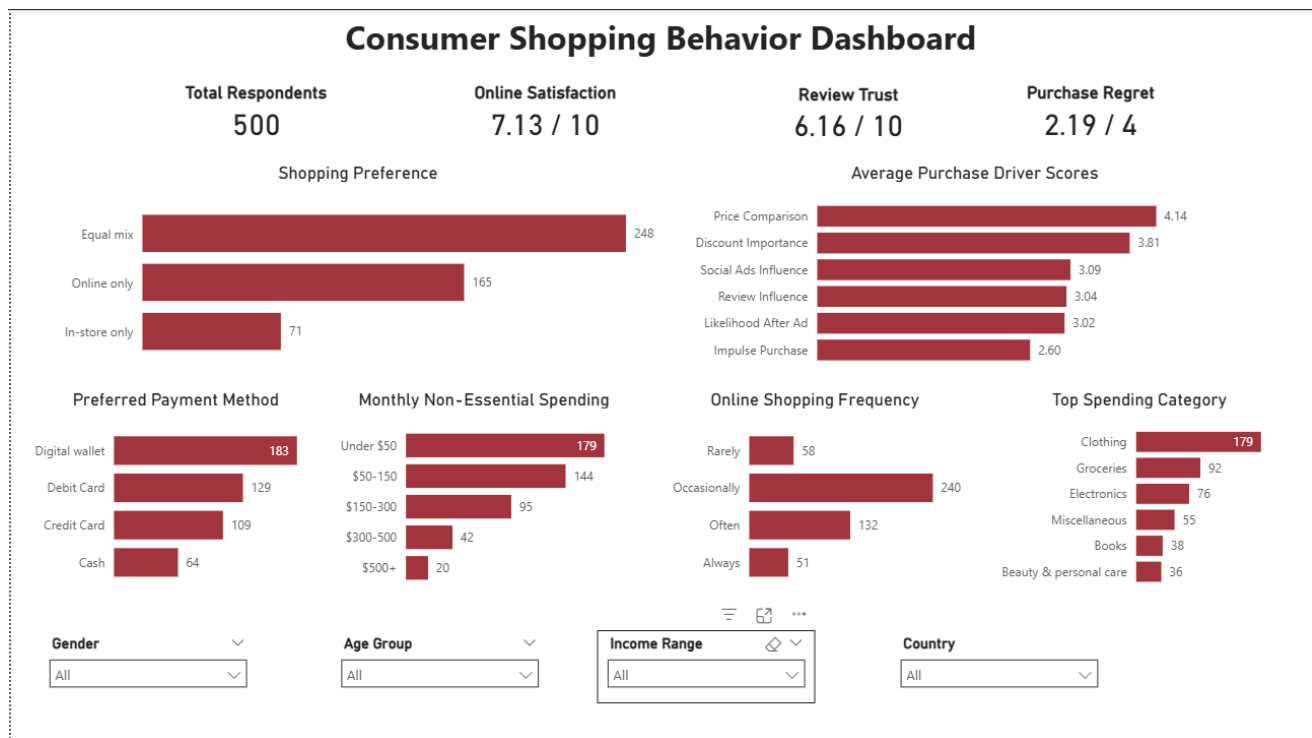


Figure 1. Final Consumer Shopping Behavior Dashboard

Dashboard Element	Purpose
KPI Cards	Total respondents, online satisfaction, review trust, and purchase regret
Shopping Preference	Compares equal-mix, online-only, and in-store-only preferences
Purchase Driver Scores	Ranks average reported purchase-driver scores
Preferred Payment Method	Shows payment preferences among respondents
Monthly Non-Essential Spending	Shows distribution across spending bands
Online Shopping Frequency	Shows how often respondents shop online
Top Spending Categories	Highlights the most frequently reported top-spending categories
Slicers	Gender, Age Group, Income Range, and Country

7. Key Insights

Omnichannel preference is prominent: 248 of 500 respondents (49.6%) prefer an equal mix of online and in-store shopping. Online-only accounts for 165 respondents (33.0%), while 71 (14.2%) prefer in-store only.

Price-conscious behavior stands out: Price Comparison is the highest average purchase-driver score at 4.14/5, followed by Discount Importance at 3.81/5. Social Ads Influence (3.09/5), Review Influence (3.04/5), and Likelihood After Ad (3.02/5) are closer to the midpoint, while Impulse Purchase is lower at 2.60/5.

Online shopping is common but often occasional: 240 respondents (48.0%) report shopping online occasionally. 132 report Often, 51 Always, and 58 Rarely.

Digital wallets lead payment preference: Digital wallets are the most frequently selected preferred payment method (183 respondents), ahead of debit cards (129), credit cards (109), and cash (64).

Non-essential spending is concentrated in lower bands: 179 respondents report spending under \$50 per month and 144 report \$50–150. Higher spending bands contain progressively fewer respondents.

Clothing is the leading top-spending category: Clothing is selected by 179 respondents as their top spending category, ahead of groceries (92) and electronics (76).

Online satisfaction is positive, while review trust is more moderate: Average online-shopping satisfaction is 7.13/10, while average trust in online reviews is 6.16/10. Average purchase-regret score is 2.19/4.

8. Business Recommendations

Strengthen omnichannel experiences: Because the largest group prefers a mix of online and in-store shopping, businesses could improve consistency across channels through aligned pricing, inventory visibility, and flexible fulfillment options.

Compete clearly on value: The high price-comparison and discount-importance scores suggest that transparent pricing, relevant promotions, and value communication may resonate with this survey audience.

Nurture occasional online shoppers: Since occasional shoppers form the largest online-frequency group, businesses could test loyalty programs, personalized offers, simplified checkout, and other retention tactics to encourage repeat digital engagement.

Prioritize convenient digital payments: Digital wallets are the leading reported payment preference, so businesses serving a similar audience should ensure reliable wallet support and smooth mobile checkout while retaining card and cash alternatives.

Segment offers by spending band: Lower-to-mid spending bands dominate the sample. Entry-level offers and bundles could be tested for these segments, while higher-spending groups may warrant separate premium strategies.

Use category insights carefully: Clothing is the most frequently reported top-spending category in this survey. Retailers in comparable markets could test stronger merchandising or promotions in this category, but should validate decisions against actual sales data.

Improve review credibility: Because review trust is lower than overall online-shopping satisfaction, businesses could strengthen review transparency with verified-purchase labels, clear moderation policies, and richer review information.

9. Tools & Technologies

Tool / Skill	How It Was Used
Power BI	Interactive dashboard creation, visual analysis, slicers, KPI cards, and report design
Power Query	Data cleaning, type conversion, null handling, text cleaning, helper columns, and transformations
DAX	Measures such as Total Respondents, average satisfaction, average review trust, average purchase regret, and purchase-driver averages
Data Visualization	Bar/column charts, 100% stacked charts, KPI cards, sorting, labeling, and dashboard layout
Exploratory Data Analysis	Distribution analysis, grouped averages, relationship exploration, and interpretation
Business Analysis	Translated analytical findings into business-oriented insights and recommendations

10. Limitations

- The dataset is survey-based and relies on self-reported behavior, which may differ from actual transaction behavior.

- The sample contains 500 respondents; findings should not automatically be generalized to a wider population without understanding the sampling method and population coverage.
- Missing responses exist across several variables. Categorical missing values were retained transparently as 'Unknown' where appropriate, while missing numeric scores remained null.
- Income and monthly non-essential spending are provided as ranges rather than exact values, so exact average income or spending cannot be calculated without making assumptions.
- The analysis identifies associations and descriptive patterns, not causal relationships.
- The meaning of the 1–4 purchase-regret scale was not independently relabeled beyond the numeric score because the exact category wording was not confirmed.
- Dashboard recommendations are directional and should be validated with business-specific sales, profitability, customer, and experimental data before implementation.

11. Conclusion

This project demonstrates an end-to-end Power BI analysis workflow: understanding a survey dataset, cleaning and transforming data in Power Query, creating DAX measures, conducting exploratory analysis, testing selected relationships, and designing an interactive dashboard.

The analysis indicates that the surveyed consumers are strongly oriented toward mixed online/in-store shopping, price comparison and discounts are prominent reported purchase drivers, digital wallets are the leading payment preference, and lower non-essential spending bands contain the largest respondent groups. Online-shopping satisfaction is positive overall, while trust in online reviews is comparatively more moderate.

The project also illustrates an important analytical principle: not every plausible relationship is supported by the data. Several EDA comparisons were relatively flat, reinforcing the need to report what the data shows rather than forcing a desired conclusion.